

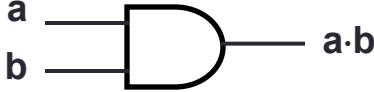
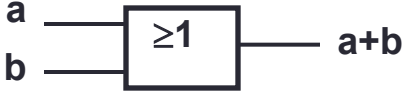
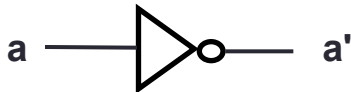
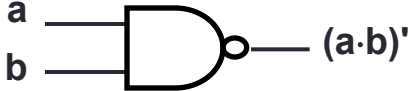
Lecture #14: Logic Circuits

1. Logic Gates
 - 1.1 Inverter/AND/OR Gates
 - 1.2 NAND/NOR Gates
 - 1.3 XOR/XNOR Gates
2. Logic Circuits
 - 2.1 Drawing and Analysing Logic Circuits
3. Universal Gates
 - 3.1 NAND Gate
 - 3.2 NOR Gate
 - 3.3 SOP and NAND Circuits
 - 3.4 POS and NOR Circuits
4. Integrated Circuit (IC) Chip
5. Programmable Logic Array
6. Read Only Memory (ROM)



1. Logic Gates

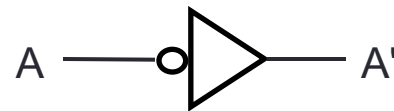
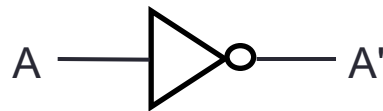
Gate symbols

	Symbol set 1	Symbol set 2 (ANSI/IEEE Standard 91-1984)
AND		
OR		
NOT		
NAND		
NOR		
EXCLUSIVE OR		



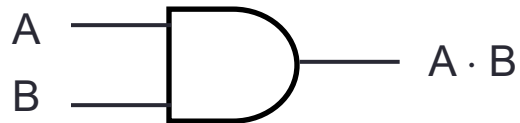
1.1 Inverter/AND/OR Gates

■ Inverter (NOT gate)



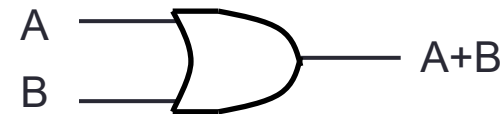
A	A'
0	1
1	0

■ AND gate n, NUS



A	B	A · B
0	0	0
0	1	0
1	0	0
1	1	1

■ OR gate

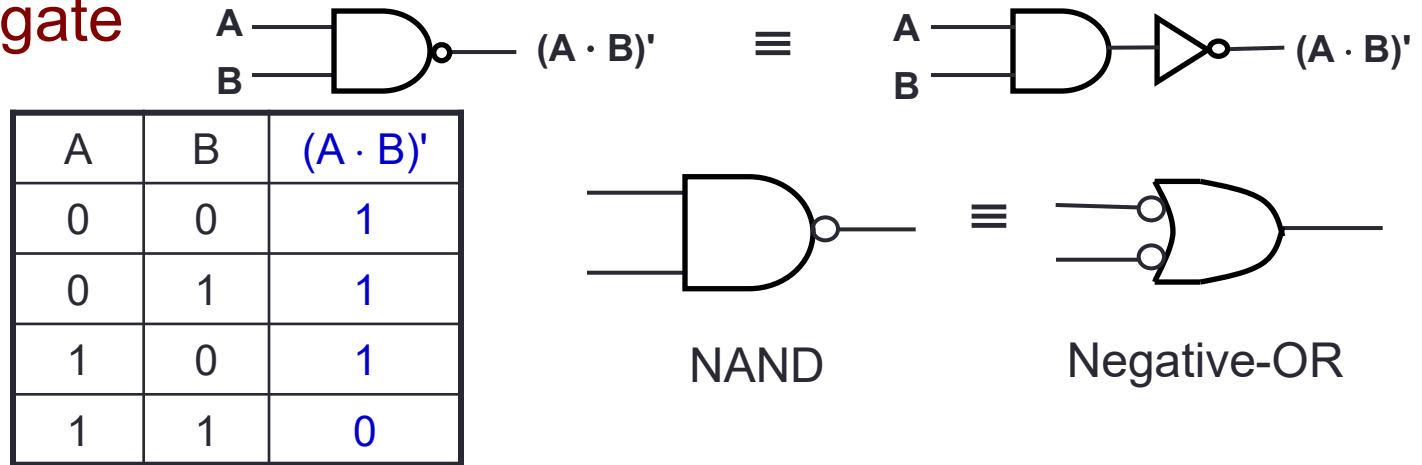


A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

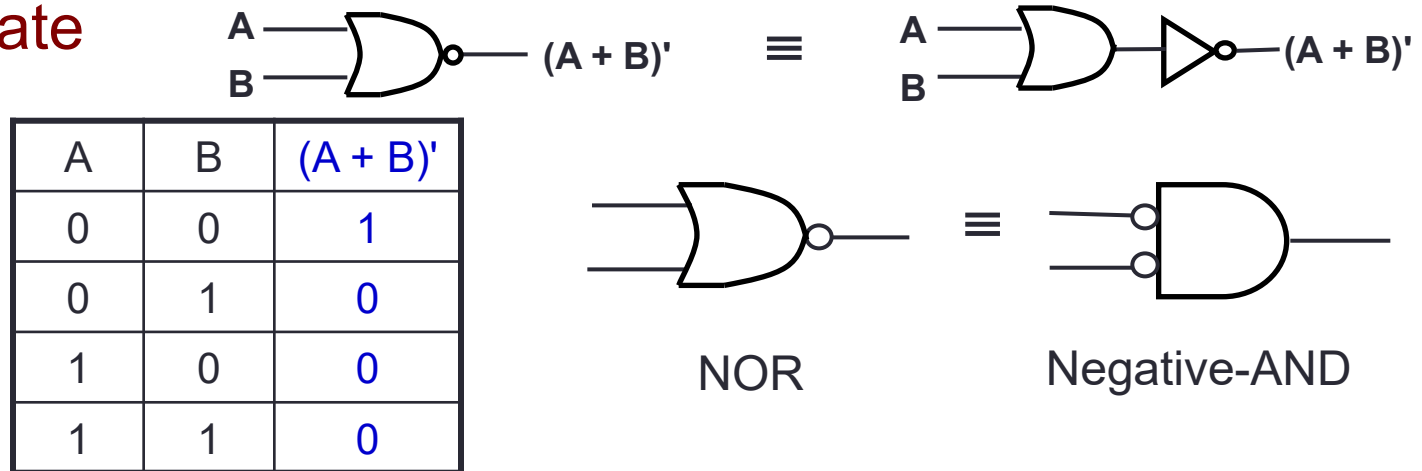


1.2 NAND/NOR Gates

■ NAND gate



■ NOR gate



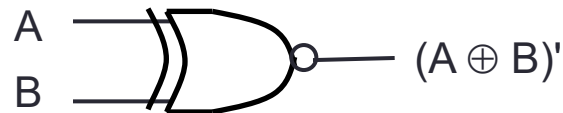
1.3 XOR/XNOR Gates

■ XOR gate



A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

■ XNOR gate



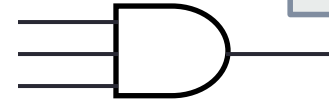
A	B	$(A \oplus B)'$
0	0	1
0	1	0
1	0	0
1	1	1

XNOR can be represented by $\odot$
(Example: $A \odot B$)



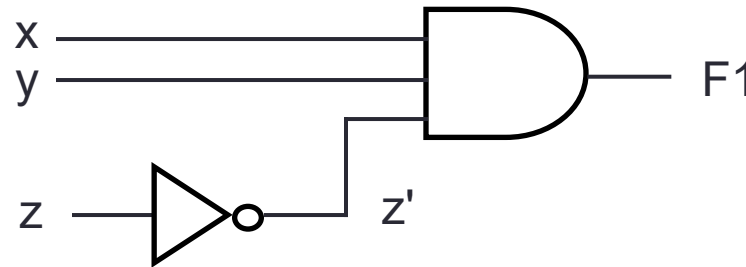
2. Logic Circuits (1/2)

- **Fan-in:** the number of inputs of a gate.
- Gates may have fan-in more than 2.
 - Example: a 3-input AND gate



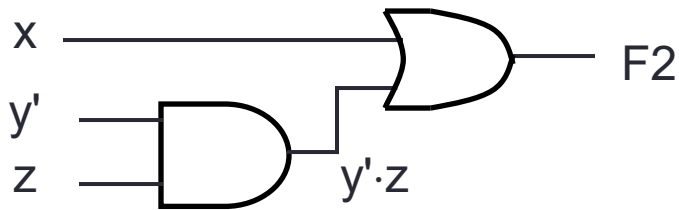
Every input must be connected in a working circuit!

- Given a Boolean expression, we may implement it as a logic circuit.
- Example: $F1 = x \cdot y \cdot z'$ (note the use of a 3-input AND gate)

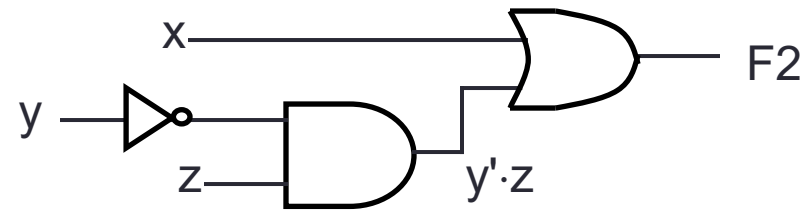


2. Logic Circuits (2/2)

- Example: $F2 = x + y' \cdot z$

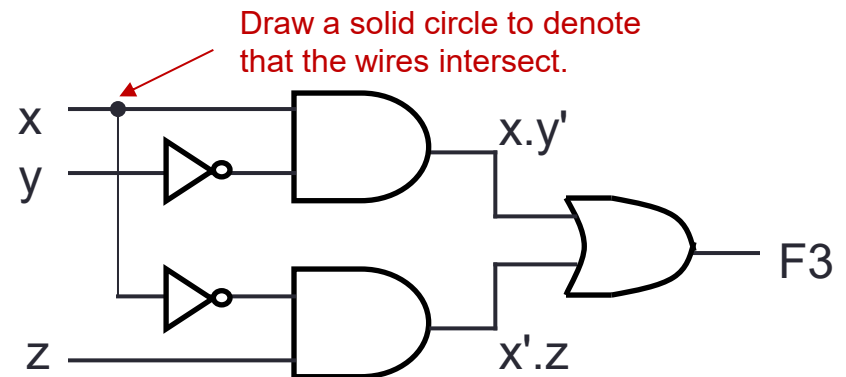
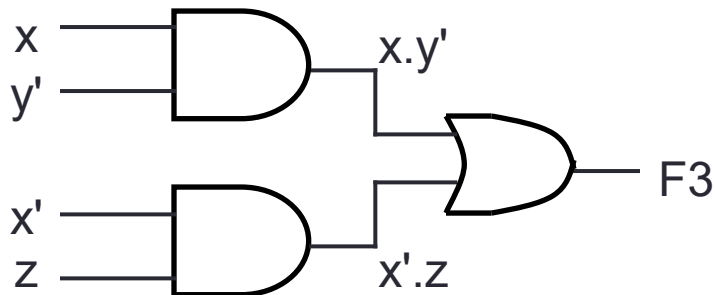


*If complemented literals
are available*



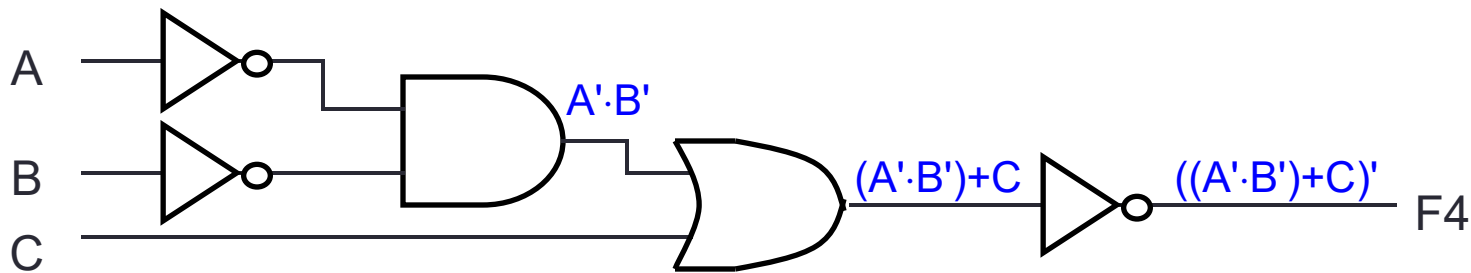
*If complemented literals
are not available*

- Example: $F3 = x \cdot y' + x' \cdot z$



2.1 Analysing Logic Circuits

- Given a logic circuit, we can analyse it to obtain the logic expression.
- Example: Given the logic circuit below, what is the Boolean expression of F4?



$$F4 = ((A'B') + C)' = (A + B) \cdot C'$$

- DLD page79 Quick Review Questions Questions 4-1 to 4-4.



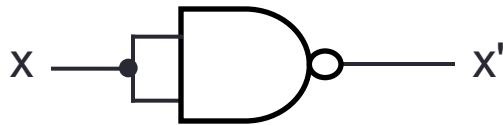
3. Universal Gates

- AND/OR/NOT gates are sufficient for building any Boolean function.
- We call the set {AND, OR, NOT} a **complete set of logic**.
- However, other gates are also used:
 - Usefulness (eg: XOR gate for parity bit generation)
 - Economical
 - Self-sufficient (eg: NAND/NOR gates)

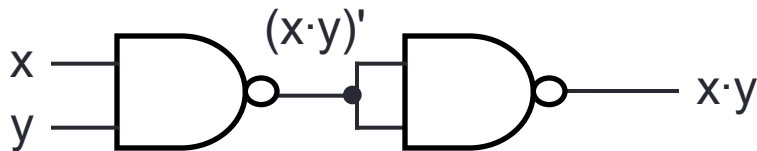


3.1 Universal Gates: NAND Gate

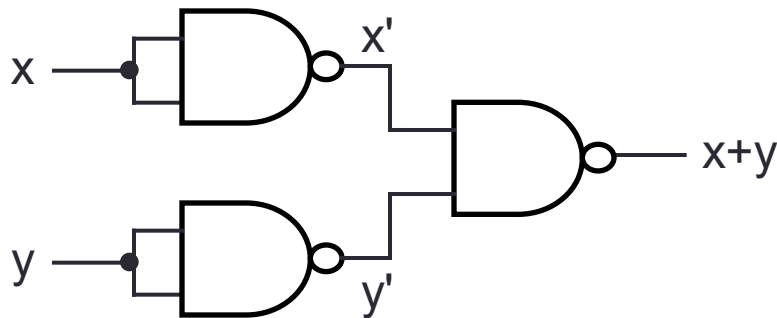
- **{NAND}** is a complete set of logic.
- Proof: Implement NOT/AND/OR using only NAND gates.



$$(x \cdot x)' = x' \quad (\text{idempotency})$$



$$\begin{aligned} ((x \cdot y)' \cdot (x \cdot y)')' &= ((x \cdot y)')' && (\text{idempotency}) \\ &= x \cdot y && (\text{involution}) \end{aligned}$$

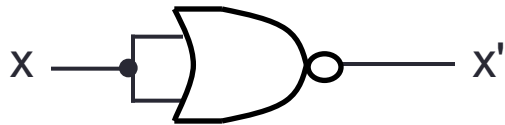


$$\begin{aligned} ((x \cdot x)' \cdot (y \cdot y)')' &= (x' \cdot y')' && (\text{idempotency}) \\ &= (x')' + (y')' && (\text{DeMorgan}) \\ &= x + y && (\text{involution}) \end{aligned}$$

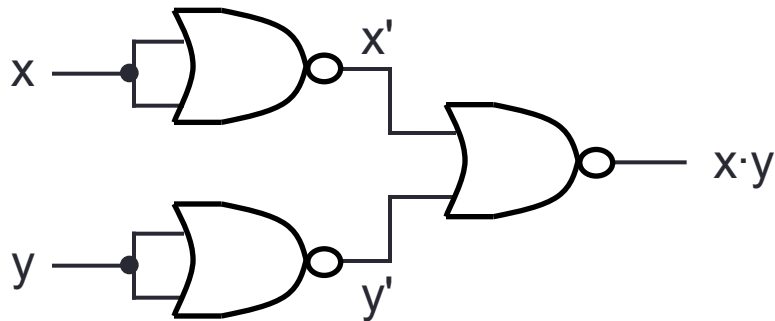


3.2 Universal Gates: NOR Gate

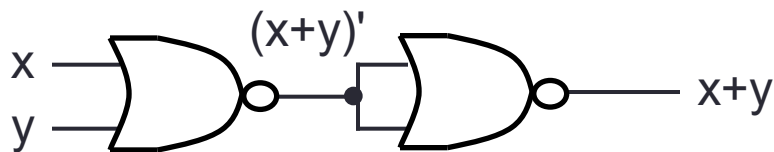
- {NOR} is a complete set of logic.
- Proof: Implement NOT/AND/OR using only NOR gates.



$$(x+x)' = x' \quad (\text{idempotency})$$



$$\begin{aligned} ((x+x)' + (y+y'))' &= (x' + y')' && (\text{idempotency}) \\ &= (x')' \cdot (y')' && (\text{DeMorgan}) \\ &= x \cdot y && (\text{involution}) \end{aligned}$$



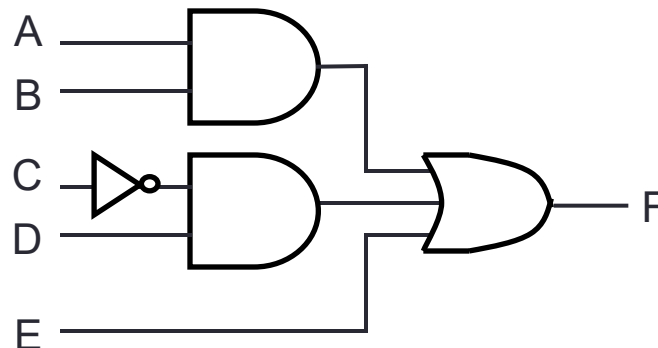
$$\begin{aligned} ((x+y)' + (x+y))' &= ((x+y))' && (\text{idempotency}) \\ &= x+y && (\text{involution}) \end{aligned}$$

- DLD page79 Quick Review Questions Questions 4-6 to 4-8.



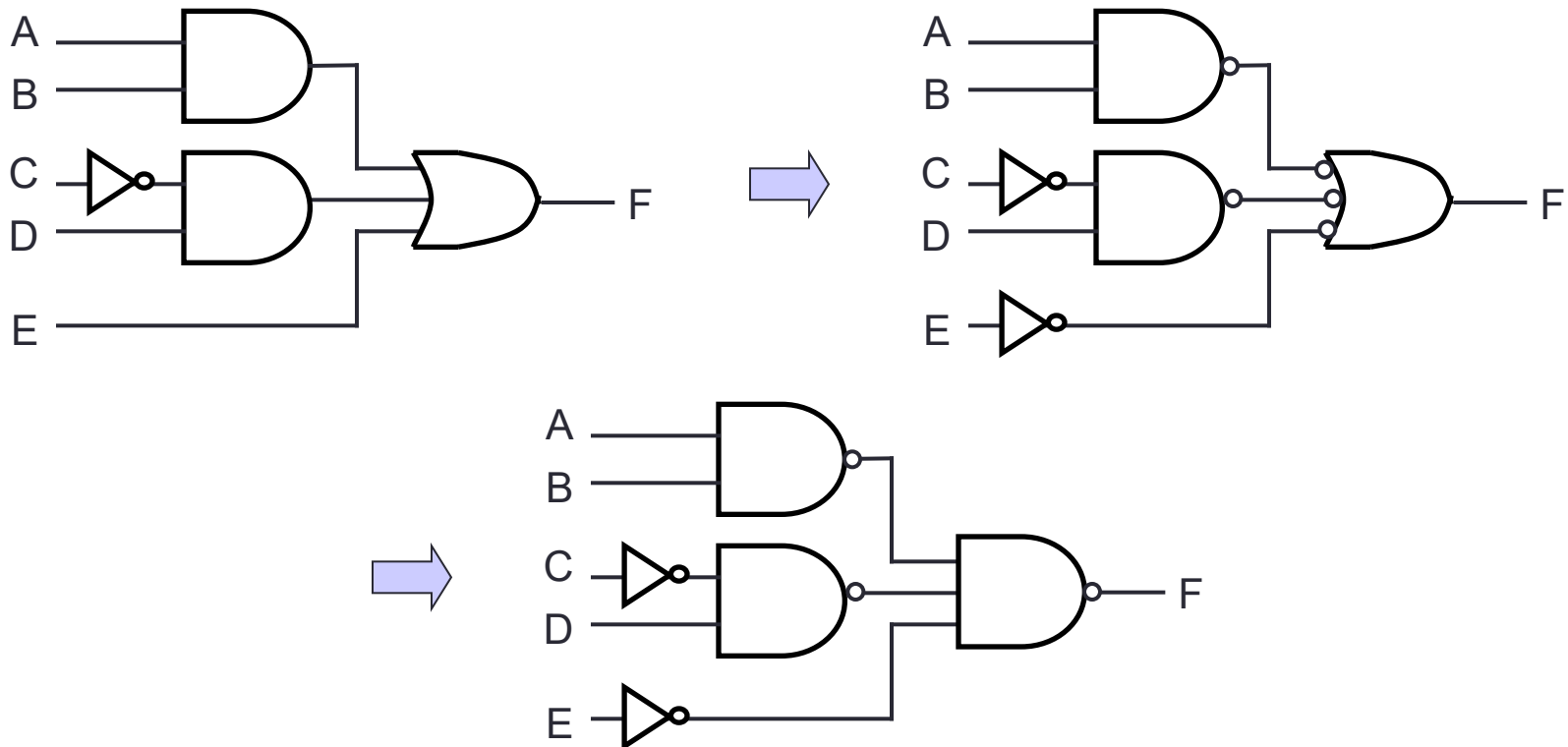
3.3 SOP and NAND Circuits (1/2)

- An SOP expression can be easily implemented using
 - 2-level AND-OR circuit
 - 2-level NAND circuit
- Example: $F = A \cdot B + C' \cdot D + E$
 - Using 2-level AND-OR circuit



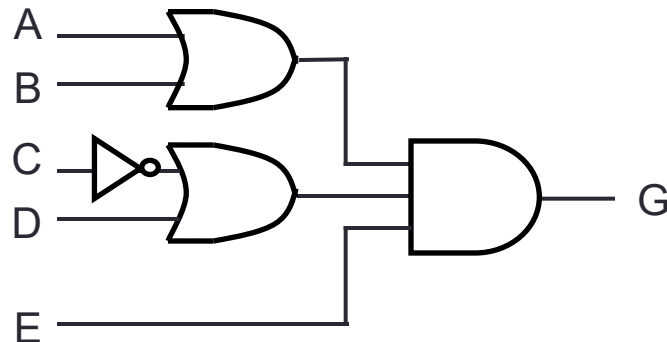
3.3 SOP and NAND Circuits (2/2)

- Example: $F = A \cdot B + C' \cdot D + E$
 - Using 2-level NAND circuit



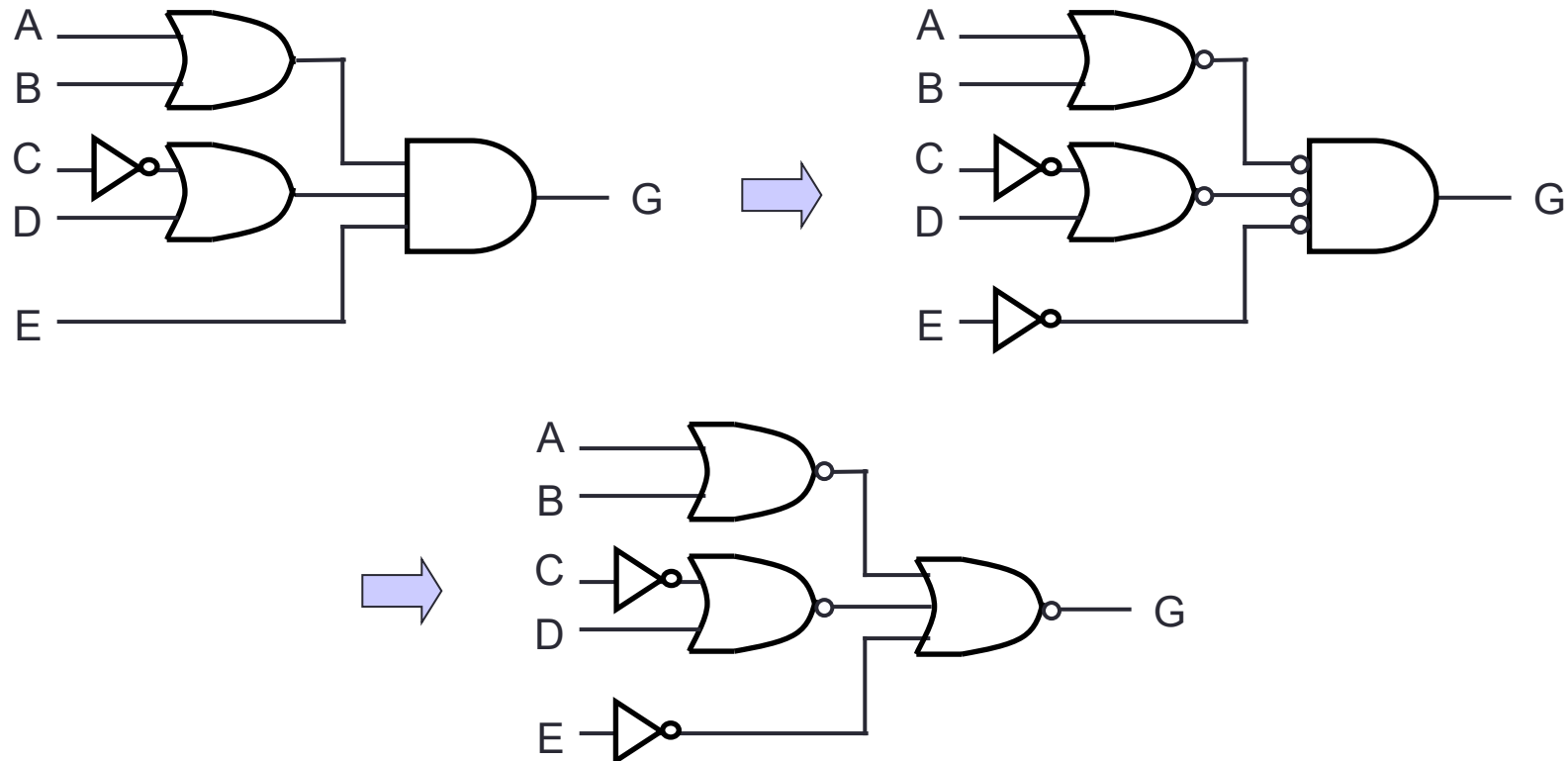
3.4 POS and NOR Circuits (1/2)

- A POS expression can be easily implemented using
 - 2-level OR-AND circuit
 - 2-level NOR circuit
- Example: $G = (A+B) \cdot (C'+D) \cdot E$
 - Using 2-level OR-AND circuit

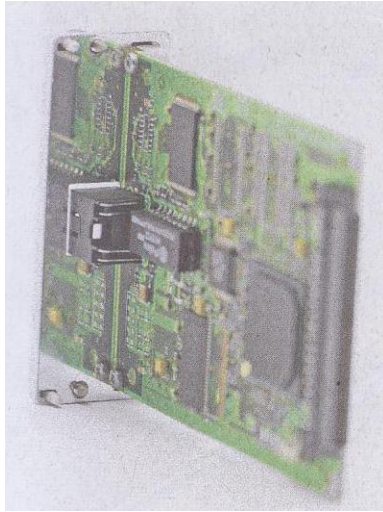


3.4 POS and NOR Circuits (2/2)

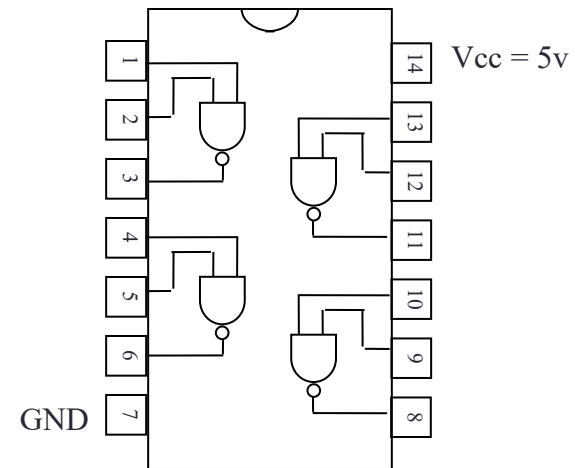
- Example: $G = (A+B) \cdot (C'+D) \cdot E$
 - Using 2-level NOR circuit



4. Integrated Circuit (IC) Chip

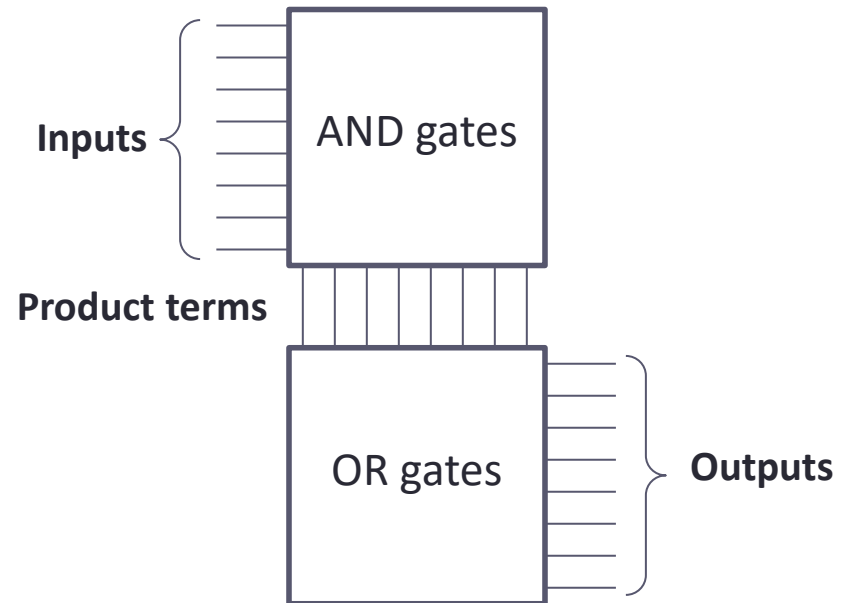


- Example of a **74LS00** chip: Quad NAND gates.



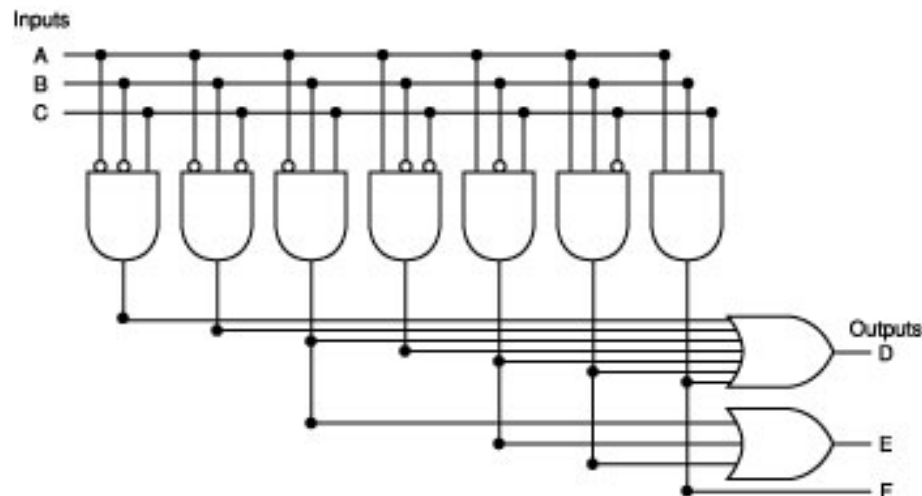
5. Programming Logic Array (PLA) (1/3)

- A programmable integrated circuit – implements sum-of-products circuits (allow multiple outputs).
- **2 stages**
 - AND gates = product terms
 - OR gates = outputs
- Connections between inputs and the planes can be 'burned'.

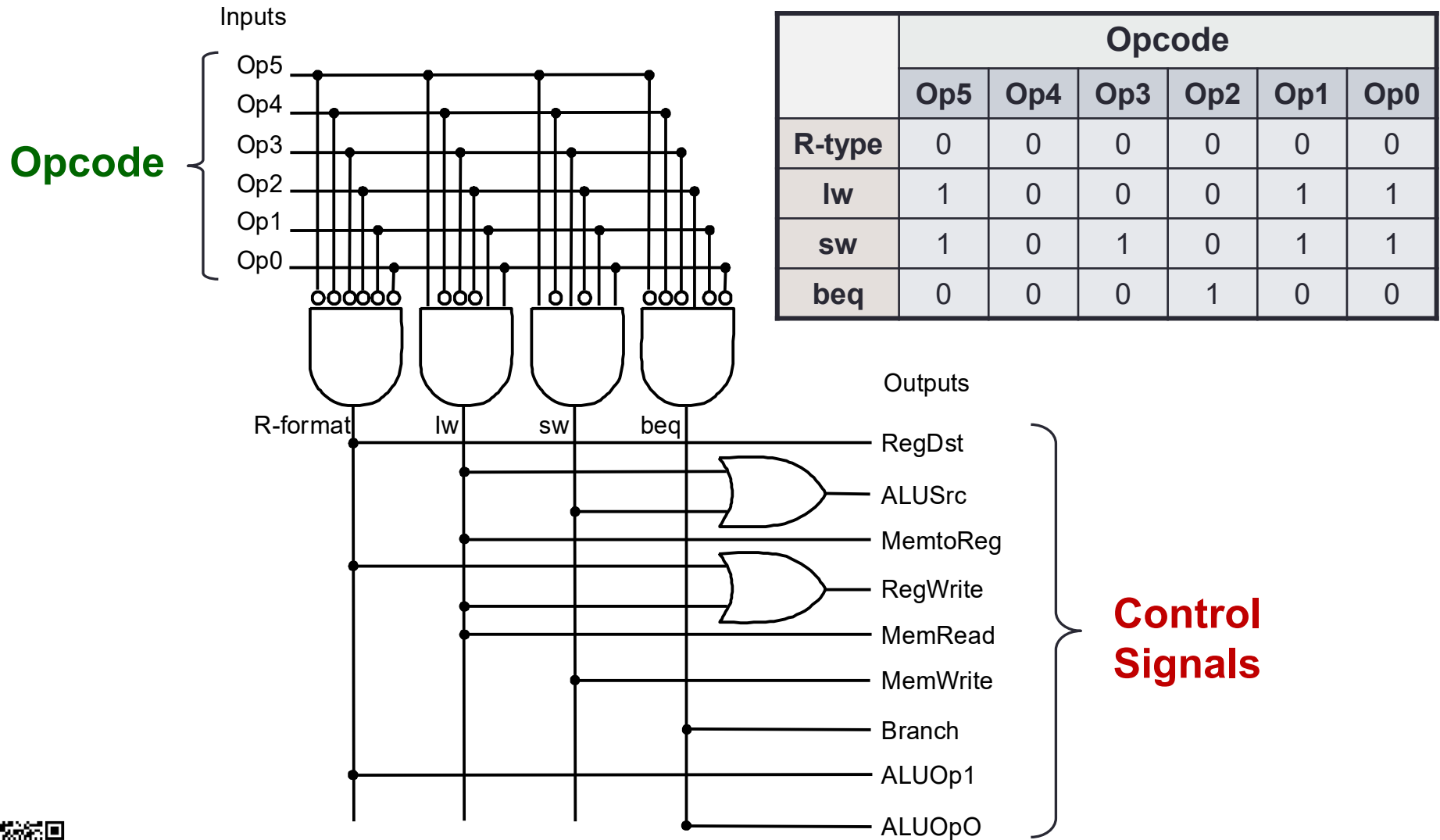


5. PLA Example (2/3)

Inputs			Outputs		
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	0	1

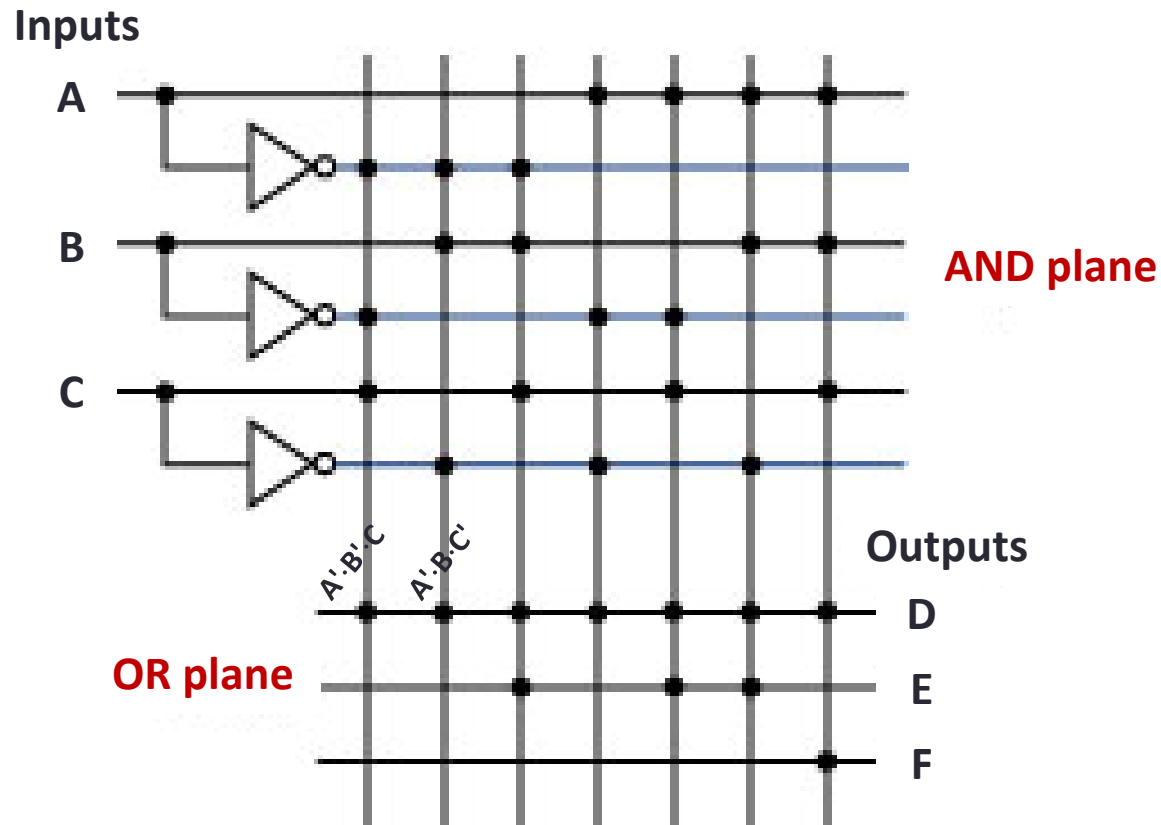


5. Combinational Circuit Implementation



5. PLA Example (3/3)

- Simplified representation of previous PLA.



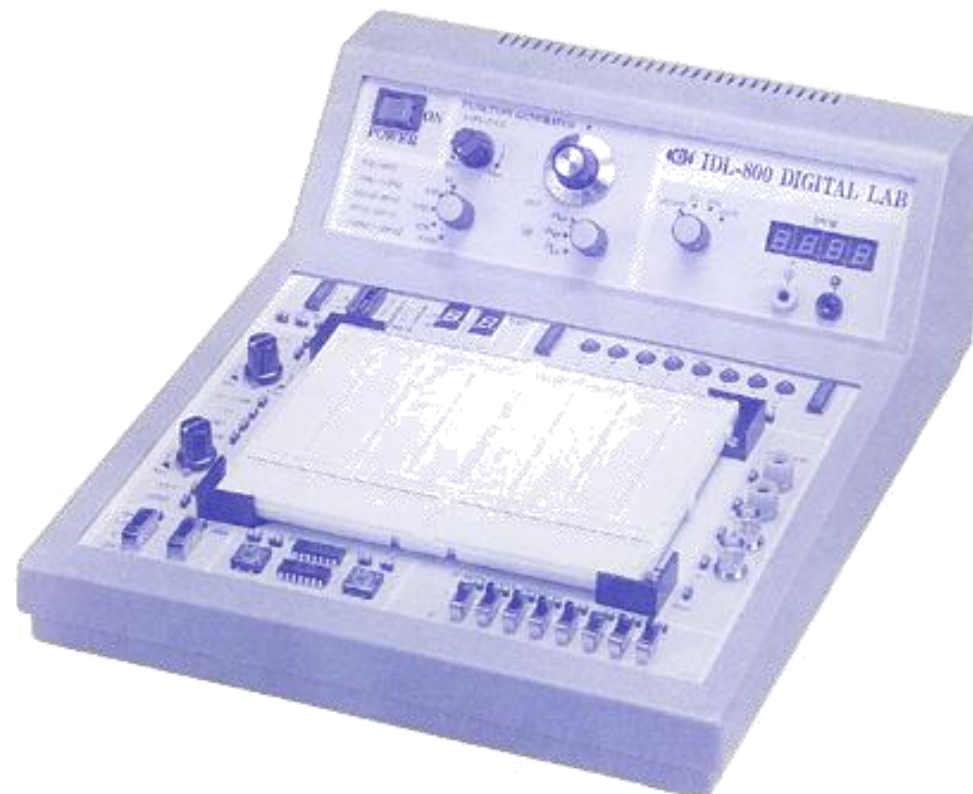
6. Read Only Memory (ROM)

- Similar to PLA
 - Set of inputs (called addresses)
 - Set of outputs
 - Programmable mapping between inputs and outputs
- Fully decoded: able to implement any mapping.
- In contrast, PLAs may not be able to implement a given mapping due to not having enough minterms.



Lab Assignments (1/2)

- For the next few labs, you will implement simple circuits using the Logic Trainer



End of File

